

高等数学习题册 第十章 重积分

习题 10.3 & 10.4 完整解答

内容：三重积分计算 + 重积分应用
(曲面面积、质心、转动惯量)

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习题 10.3 三重积分

第1题 化三重积分为三次积分

(1) Ω 由 $z=xy$, $z=0$, $x+y=2$ ($x \geq 0, y \geq 0$) 围成。

【解】D: $0 \leq x \leq 2, 0 \leq y \leq 2-x; z: 0 \leq z \leq xy$

$$I = \int_0^2 dx \int_0^{2-x} dy \int_0^{xy} f(x, y, z) dz$$

(2) Ω 由 $z=x+y$ 及 $z=4$ 围成。

【解】D: $x+y \leq 4; z: x+y \leq z \leq 4$

$$I = \int_0^4 dx \int_{-\sqrt{4-x}}^{\sqrt{4-x}} dy \int_{x+y}^4 f(x, y, z) dz$$

第2题 计算三重积分

【题】 $I = \iiint_{\Omega} x\sqrt{y}z \, dV$, $\Omega: z=xy, z=0, x=2, y=x$ ($y \geq 0$)

【解】区域: $0 \leq x \leq 2, 0 \leq y \leq x, 0 \leq z \leq xy$

先对 z : $\int_0^{xy} (xy) z \, dz = (xy) \frac{z^2}{2} \Big|_0^{xy} = x\sqrt{y} \frac{y}{2}$

$$\begin{aligned} I &= (1/6) \int_0^2 \int_0^x x\sqrt{y} \, dy \, dx \\ &= (1/6) \int_0^2 x \frac{2}{3} (x\sqrt{8}) \, dx = (1/48) \int_0^2 x^3 \, dx \\ &= (1/48) \cdot (2^4/4) = 131072/816 = 8192/51 \end{aligned}$$

答: $I = 8192/51$

第3题 球面坐标

【题】 $I = \iiint_{\Omega} x\sqrt{y}z \, dV$, $\Omega: x+y+z \leq 2$, 第一卦限

【解】令 $x=r\sin\phi\cos\theta, y=r\sin\phi\sin\theta, z=r\cos\phi$

$0 \leq r \leq \sqrt{2}, 0 \leq \phi \leq \pi/2, 0 \leq \theta \leq \pi/2$

被积函数 $= r\sin\phi\cos\theta \cos\phi \cos\theta \sin\theta \cdot \cos\phi$

$$I = \int_0^{\sqrt{2}} r \, dr \cdot \int_0^{\pi/2} \sin\phi \cos\phi \, d\phi \cdot \int_0^{\pi/2} \cos^2\theta \sin\theta \, d\theta$$

$$\textcircled{1} r \, dr: \left[\frac{r^2}{2} \right]_0^{\sqrt{2}} = 16/8 = 2$$

$$\textcircled{2} \sin\phi \cos\phi \, d\phi: \left[\frac{\sin^2\phi}{2} \right]_0^{\pi/2} = 1/6$$

$$\textcircled{3} \cos^2\theta \sin\theta \, d\theta: \left[-\cos^3\theta/3 \right]_0^{\pi/2} = 1/4$$

$$I = 2 \times (1/6) \times (1/4) = 1/12$$

答: $I = 1/12$

第4题 锥面区域

【题】 $I = \iiint_{\Omega} z \, dV$, $\Omega: z=(h/R)\sqrt{x^2+y^2}$ 到 $z=h$

【解】柱面坐标: $0 \leq \theta \leq 2\pi, 0 \leq \rho \leq R, h\rho/R \leq z \leq h$

$$\int_{h\rho/R}^h z \, dz = h \frac{z^2}{2} \Big|_{h\rho/R}^h = (h^2/2)(1 - \rho^2/R^2)$$

$$\begin{aligned} I &= 2\pi \cdot (h^2/2) \int_0^R \rho(1 - \rho^2/R^2) \, d\rho \\ &= \pi h^2 \left[\rho^2/2 - \rho^4/(4R^2) \right]_0^R = \pi h^2 (R^2/2 - R^4/4R^2) = \pi R^2 h^2/4 \end{aligned}$$

答: $I = \pi R^2 h^2/4$

第5题 柱面坐标

(1) $I = \iiint_{\Omega} z \, dV$, Ω : $z=2-\rho$ 与 $z=\rho$ 围成

【解】交线: $2-\rho = \rho \Rightarrow \rho = 1$

$$\int_0^{2\pi} \int_0^1 \int_{2-\rho}^{\rho} z \, dz \, d\rho \, d\theta = [(2-\rho)^2 - \rho^2] / 2 \Big|_0^1 = (4-4\rho) / 2 = 2-2\rho$$

$$I = 2\pi \int_0^1 (2-2\rho) \rho \, d\rho = 4\pi \left[\rho - \frac{2}{3}\rho^3 \right]_0^1 = 4\pi (1/3) = \frac{4}{3}\pi$$

答: $I = \frac{4}{3}\pi$

(2) $I = \iiint_{\Omega} (x+y+z) \, dV$, Ω : $\rho \leq 4z$ 到 $z=4$

【解】 $z = \rho/4$ 到 $z=4$, $\rho \leq 4$

$$I = 2\pi \int_0^4 \int_0^{4-z} \int_z^4 (x+y+z) \, dz \, d\rho \, d\theta$$

$$= 2\pi (256 - 4096/24) = 2\pi \cdot (256/3) = 512\pi/3$$

答: $I = 512\pi/3$

第6题 球面坐标

(1) $I = \iiint_{\Omega} (x+y+z) \, dV$, Ω : $r \leq 2$

【解】 $I = \int_0^{2\pi} \int_0^\pi \int_0^2 r^3 \sin\phi \, dr \, d\phi \, d\theta$

$$= 2\pi \cdot 2 \cdot \left[\frac{r^4}{4} \right]_0^2 = 4\pi \cdot (32/5) = 128\pi/5$$

答: $I = 128\pi/5$

(2) $I = \iiint_{\Omega} z \, dV$, Ω : $r \leq 8\cos\phi$, $\phi \leq \pi/4$

【解】 $I = 2\pi \int_0^{\pi/4} \int_0^{8\cos\phi} \int_0^{8\cos\phi} z \, dz \, d\phi \, d\theta$

$$= (2\pi \cdot 32768/5) \int_0^{\pi/4} \cos^3\phi \sin\phi \, d\phi$$

$$= (2\pi \cdot 32768/5) \cdot [-\cos^4\phi / 4]_0^{\pi/4}$$

$$= (2\pi \cdot 32768/5) \cdot (1 - 1/16) / 4 = 1536\pi$$

答: $I = 1536\pi$

第7题 选用适当坐标

(1) $I = \iiint_{\Omega} (x+y) dV$, $\Omega: x+y \leq 4, 0 \leq z \leq 2$, 第一卦限

【解】柱面坐标: $x = \rho \cos \theta, y = \rho \sin \theta$

$$\begin{aligned} I &= \int_0^2 \int_0^{\pi/2} \int_0^{4-\rho \sin \theta} (\rho \cos \theta + \rho \sin \theta) \rho d\rho d\theta dz \\ &= [\cos \theta / 3]_0^{\pi/2} \cdot [\rho^2 / 2]_0^{4-\rho \sin \theta} \cdot 2 \\ &= (1/3) \cdot (32/5) \cdot 2 = 64/15 \end{aligned}$$

答: $I = 64/15$

(2) $I = \iiint_{\Omega} (x+y) dV$, $\Omega: z = 9 - \rho^2$ 到 $z = 4$

【解】 $\rho \leq 8/3$

$$I = 2\pi \int_0^{8/3} \rho (4 - 9\rho^2) d\rho = 4096\pi/243$$

答: $I = 4096\pi/243$

(3) $I = \iiint_{\Omega} \sqrt{x+y+z} dV$, $\Omega: x+y+z \leq 2z$

【解】球面 $r = 2\cos \phi$ ($0 \leq \phi \leq \pi/2$)

$$\begin{aligned} I &= 2\pi \int_0^{\pi/2} \sin \phi \cdot [r^4/4]_0^{2\cos \phi} d\phi \\ &= 8\pi \int_0^{\pi/2} \cos^4 \phi \sin \phi d\phi = 8\pi \cdot [\cos^5 \phi / 5]_0^{\pi/2} = 8\pi/5 \end{aligned}$$

答: $I = 8\pi/5$

第8题 求体积

(1) $z = 6 - x^2 - y^2$ 与 $z = x^2 + y^2$ 围成的体积

【解】交线: $6 - \rho^2 = \rho^2 \Rightarrow \rho = \sqrt{3}$

$$\begin{aligned} V &= 2\pi \int_0^{\sqrt{3}} \rho (6 - 2\rho^2) d\rho = 2\pi [3\rho^2 - \rho^4/2]_0^{\sqrt{3}} \\ &= 2\pi (9 - 9/2) = 9\pi \end{aligned}$$

答: $V = 9\pi$

(2) $x^2 + y^2 + z^2 = 6z$ 与 $x^2 + y^2 = z^2$ (含 z 轴)

【解】球坐标: 球面 $r = 6\cos \phi$, 锥面 $\phi = \pi/4$

$$\begin{aligned} V &= 2\pi \int_0^{\pi/4} \sin \phi \cdot (6\cos \phi)^3 d\phi \\ &= 144\pi \int_0^{\pi/4} \cos^3 \phi \sin \phi d\phi \\ &= 144\pi \cdot [-\cos^4 \phi / 4]_0^{\pi/4} = 144\pi \cdot (1 - 1/4)/4 = 27\pi \end{aligned}$$

答: $V = 27\pi$

习题 10.4 重积分的应用

第1题 曲面面积

【题】球面 $x^2+y^2+z^2=a^2$ 含在柱面 $x^2+y^2=ax$ 内的面积

【解】上半球面 $z=\sqrt{a^2-x^2-y^2}$

$$z/x=-x/z, \quad z/y=-y/z$$

$$\sqrt{1+z_x^2+z_y^2} = a/\sqrt{a^2-x^2-y^2} = a/\sqrt{a^2-\rho^2}$$

投影 D: $\rho \leq a \cos \theta, \quad \theta \in [-\pi/2, \pi/2]$

$$\begin{aligned} S_{\text{上}} &= \int_{-\pi/2}^{\pi/2} d\theta \int_0^{a \cos \theta} a \cdot \rho / \sqrt{a^2-\rho^2} d\rho \\ &= \int_{-\pi/2}^{\pi/2} a[-\sqrt{a^2-\rho^2}]_0^{a \cos \theta} d\theta \\ &= a \int_{-\pi/2}^{\pi/2} (1-|\sin \theta|) d\theta = a(\pi-2) \end{aligned}$$

总面积 $S = 2a(\pi-2)$

答: $S = 2a(\pi-2)$

第2题 曲面面积

【题】锥面 $z=\sqrt{x^2+y^2}$ 被柱面 $x^2+y^2=2x$ 割下的面积

【解】 $z=\rho \rightarrow z_x=x/\rho, \quad z_y=y/\rho$

$$\sqrt{1+z_x^2+z_y^2} = \sqrt{1+1} = \sqrt{2}$$

D: $\rho \leq 2 \cos \theta, \quad \theta \in [-\pi/2, \pi/2]$

$$\begin{aligned} S &= \sqrt{2} \cdot \int_{-\pi/2}^{\pi/2} d\theta \int_0^{2 \cos \theta} \rho d\rho \\ &= \sqrt{2} \cdot \int_{-\pi/2}^{\pi/2} 2 \cos^2 \theta d\theta = \sqrt{2} \cdot \pi \end{aligned}$$

答: $S = \sqrt{2} \cdot \pi$

第3题 质心

【题】D 由 $y=\sqrt{x}$ 及 $y=x$ 围成, 面密度 $\rho(x,y)=x$, 求质心

【解】交点 $(0,0), (1,1)$, 用 $0 \leq y \leq 1, y \leq x \leq y^2$

$$\begin{aligned} M &= \int_0^1 dy \int_y^{y^2} x dx = (1/3) \int_0^1 (y^3 - y) dy \\ &= (1/3) (1/4 - 1/2) = (1/3) (3/28) = 1/28 \end{aligned}$$

$$\begin{aligned} M_x &= \int_0^1 dy \int_y^{y^2} y x dx = (1/4) \int_0^1 (y^3 - y) dy \\ &= (1/4) (1/5 - 1/2) = (1/4) (4/45) = 1/45 \end{aligned}$$

$$\begin{aligned} M_y &= \int_0^1 y dy \int_y^{y^2} x dx = (1/3) \int_0^1 (y^3 - y) dy \\ &= (1/3) (1/5 - 1/2) = (1/3) (3/40) = 1/40 \end{aligned}$$

$$\bar{x} = (1/45) / (1/28) = 28/45$$

$$\bar{y} = (1/40) / (1/28) = 28/40 = 7/10$$

答: 质心 $(28/45, 7/10)$

第4题 转动惯量

【题】D 由 $y=\sqrt{x}$ 与 $x=1$ 围成, 面密度=1, 求 I_x, I_y

【解】D: $0 \leq x \leq 1, -\sqrt{x}/7 \leq y \leq \sqrt{x}/7$

$$\begin{aligned} I_x &= \int_0^1 dx \int_{-\sqrt{x}/7}^{\sqrt{x}/7} y^2 dy = \int_0^1 (2/3) (\sqrt{x}/7) dx \\ &= (2/3) (1/343) \int_0^1 x^{3/2} dx = (2/1029) (2/5) = 4/5145 \end{aligned}$$

$$\begin{aligned} I_y &= \int_0^1 x^2 (2\sqrt{x}/7) dx = (2/7) \int_0^1 x^{5/2} dx \\ &= (2/7) (2/7) = 4/49 \end{aligned}$$

答: $I_x = 4/5145, I_y = 4/49$

第5题 体积、质心、转动惯量

【题】 Ω 由 $z=x+y$, $z=0$, $x=a$, $y=a$ ($x, y \geq 0$) 围成

$$\begin{aligned}(1) \quad V &= \int_0^a \int_0^{a-x} (x+y) dy dx \\ &= \int_0^a (x \cdot a + \frac{1}{2}a^2) dx = a^2 \cdot \frac{3}{2} + \frac{1}{2}a^3 = 2a^2\end{aligned}$$

答: $V = 2a^2$

$$\begin{aligned}(2) \quad &\text{由对称性 } \bar{x} = \bar{y} \\ V\bar{x} &= \int_0^a \int_0^{a-x} x(x+y) dy dx = \int_0^a (x \cdot a + \frac{1}{2}x a^2) dx \\ &= a^2 \cdot \frac{4}{3} + \frac{1}{6}a^3 = 5a^3/6 \\ \bar{x} &= (5a^3/6) / (2a^2) = 5a/8 \\ V\bar{z} &= \int_0^a \int_0^{a-x} (x+y)^2/2 dy dx \quad (\text{对 } z \text{ 积分后}) \\ &= (1/2) \int_0^a (x^2 + 2x \cdot \frac{1}{2}a + \frac{1}{3}a^2) dy dx \\ &= (1/2) (a^2 + 2a^2 + a^3) \cdot (1/a) \quad (\text{化简}) \\ \bar{z} &= 2a/5 \quad (\text{经计算}) \\ \text{答: 质心 } &(5a/8, 5a/8, 2a/5)\end{aligned}$$

$$\begin{aligned}(3) \quad I_z &= \rho \int_0^a \int_0^{a-x} (x+y)(x+y) dz dy dx \\ &= \rho \int_0^a \int_0^{a-x} (x+y)^2 dy dx \\ &= \rho \int_0^a (x^2 + 2x \cdot \frac{1}{2}a + \frac{1}{3}a^2) dy dx \\ &= \rho (a^2 + 2a^2 + a^3) \cdot a = \rho \cdot (28a^3/3) \\ \text{答: } I_z &= 28 \rho a^3/3\end{aligned}$$